
Chapter 4 System Startup

4.1 Preparation of the reagents and the standard solution



DANGER

It is dangerous to come into contact with or inhale chemicals. Only well trained and qualified persons can do the work described in this chapter.



WARNING

It is dangerous to come into contact with or inhale chemicals. For the reagents are toxic and corrosive, therefore, it is recommended to order controlled prepared reagents from HACH. With these reagents, not only personal injuries and environmental pollution can be avoided, but accurate measurement and calibration results can also be obtained.

4.2 Components of the reagents



DANGER!

The chemical reagents should be prepared by a professional staff. For sake of safety, please follow the rules below:

- Wear safety clothing (laboratory clothing)
- Wear a pair of safety goggles or a face mask
- Wear a pair of rubber gloves
- There must be an exhaust fan in the laboratory
- Use articles made of glass or teflon material only in the whole process of reagent preparation in this chapter.
- Make sure all the bottles used are ventilative after installation.
- Make sure the local applicable accident prevention regulations are followed.
- Handle relevant substances correctly in light of the local applicable laws and regulations.

4.2.1 Mercuric sulfate solution

	Basic materials	CAS No.	Quantity
A	Sulfuric acid, 95~97% ACS	7664-93-9	100 mL
B	Mercuric sulfate (II) ACS	7783-35-9	100g

The steps below are necessary for avoiding the interference caused by polluted compounds for the interference may affect the COD measurement.

1. Put 100g of substance B (mercuric sulfate (II) ACS) into a 1L measuring cup.
2. Then slowly add 800 ML of purified water into the measuring cup and stir the suspension with a magnetic stirrer for 2 hours.
3. Do suction filtration with a suction filter (sintered glass filter D1). After that, only a yellow precipitate will be left in the measuring cup.
4. Again slowly add 800 ML of distilled water into the measuring cup to carry out a flushing cycle.
5. Stir with a magnetic stirrer for 2 hours and do suction filtration with a suction filter (sintered glass filter D1). The water obtained after the suction filtration in the second flushing cycle is used for determining the COD concentration with the national standard on the COD laboratory determination method.

COD<20mg/L

1. Add 750 ML of distilled water into the precipitate (yellow alkaline mercuric sulfate) left after the second time of suction filtration slowly.
2. Stir the yellow suspension with a magnetic stirrer and add 100 ML of substance A (sulfuric acid, 95-97% p.a.) into it carefully at the same time.
3. Add purified water until the volume becomes 1 L after the mercuric sulfate is completely dissolved (the solution is clear at the same time).

COD>20mg/L

1. Add 300 ML of distilled water into the precipitate (yellow alkaline mercuric sulfate) left after the second time of suction filtration slowly.
2. Stir the yellow suspension with a magnetic stirrer and add 500 ML of substance A (sulfuric acid, 95-97% p.a.) into it carefully at the same time. After the yellow suspension is completely dissolved, there will be a white mercuric sulfate suspension.
3. Stir the white suspension in a 1L measuring cup for 24 hours.
4. Do suction filtration with a suction filter (sintered glass filter D1). After that, only a white precipitate will be left in the measuring cup.
5. Add 750 ML of purified water into the white precipitate slowly and stir the solution. When stirring is done, the white suspension will be completely dissolved and form a yellow alkaline mercuric sulfate suspension.
6. Stir continually and add 100 ML of substance A (sulfuric acid, 95-97% ACS) carefully.
7. Add purified water until the volume becomes 1 L after the mercuric sulfate is completely dissolved (the solution is clear at the same time).

4.2.2 Potassium dichromate solution

	Basic materials	CAS No.	Quantity
A	Sulfuric acid, 95~97% ACS	7664-93-9	95 mL
B	Potassium dichromate ACS	7778-50-9	80g

1. Firstly, add 700 ML of distilled water into a 1 L measuring cup.
2. Stir the distilled water with a magnetic stirrer and add 95 ML of substance A (sulfuric acid, 95-97% ACS) into it carefully at the same time.
3. Keep stirring until the solution has cooled down the ambient temperature.
4. Stir continually and add 80 g of substance B (potassium dichromate ACS) into the solution at the same time.
5. Add purified water until the volume becomes 1 L after the potassium dichromate is completely dissolved (the solution is clear at the same time).

4.2.3 Sulfuric acid

	Basic materials	CAS No.	Quantity
A	Sulfuric acid, 95~97% ACS	7664-93-9	2500 mL
B	Silver sulfate ACS	10294-26-5	40g

1. Add 2,500 ML of substance A (sulfuric acid, 95-97% ACS) into a 5 L beaker.

Important note: Never add water into the breaker!

2. Stir the solution with a magnetic stirrer and add 40g of substance B (silver sulfate ACS) into it carefully at the same time. Stir continually after that until all the substance in the beaker are completely dissolved (the process may need about 2 to 3 hours).

4.2.4 Zero-point standard solution

	Basic materials	CAS No.	Quantity
A	Distilled water	7732-18-5	500 mL
B	Sulfuric acid, 95~97% ACS	7664-93-9	0.5 mL

1. Firstly, add 400 ML of distilled water into a 500 ML measuring glass.
2. Stir the solution with a magnetic stirrer and add 0.5 ML of compound B (sulfuric acid, 95-97% ACS) into it carefully.
3. After mixing, add more distilled water until the solution reaches the scale line of 500 ML.

4.2.5 Standard solution

	Basic materials	CAS No.	Quantity
A	Potassium hydrogen phthalate	877-24-7	500 mL
B	Sulfuric acid, 95~97% ACS	7664-93-9	0.5 mL

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1. Firstly, add 200 ML of distilled water into a 250 ML measuring beaker.
 2. Stir the solution with a magnetic stirrer and add 0.31875g of substance A (potassium hydrogen phthalate) and substance B (sulfuric acid 95-97% ACS) into it carefully.
 3. Add distilled water until the volume becomes 250 ML after the potassium hydrogen phthalate is completely dissolved (the solution is clear at this time).

Remark: Before using potassium hydrogen phthalate, put it into a drying oven for baking at a temperature of 105°C.

4.3 Danger information

4.3.1 Dangerous components

- Sulfuric acid
- Mercuric sulfate
- Potassium dichromate

4.3.2 Danger identification

- Inhalation, skin contact and swallowing of the above-mentioned dangerous components may all cause poisoning.
- The above-mentioned dangerous components have a cumulative danger.
- The above-mentioned dangerous components may cause serious burns.
- The above-mentioned dangerous components are very harmful for aquatic lives and may have a long-term adverse impact on the aquatic environment.

4.3.3 First aid measures

- Summary: Take off the contaminated or wetted clothes immediately for safety treatment. Ask the opinion of a doctor immediately. If possible, show the relevant label or the list of dangerous substances.
- Inhalation: Make sure there is enough fresh air.
- Skin contact: After careless skin contact, flush skin with a lot of water immediately.
- Eye contact: After careless contact of dangerous substances with eyes, immediately flush eyes thoroughly with a lot of water and ask the opinion of a doctor.
- Swallowing: Rinse the mouth, drink a lot of water and take necessary medical first aid measures.
- Consulting with a doctor: Receive treatment in line with the symptoms.

4.3.4 Fire protection measures

- Proper extinguishing agents: The products are non-flammable; it is necessary to discuss about the extinguishing methods of the surrounding areas.
- Special exposure danger: A mercury gas may be emitted in combustion.
- Special protective articles for fire fighters: use self-provided breathing devices and